

Gaps and Overlaps When We Measure

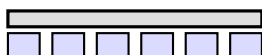
Laying units with no gaps or overlaps, spotting a gap or overlap mistake, and fixing the count

Directions: Every child on this page measured with the **same size cubes**.

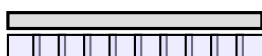
- Some children laid their cubes the **good** way. Some left **gaps**. Some let the cubes **overlap**.
- Look at the picture first. Then answer with a number, a symbol, or a drawing.
- When a question says *find the mistake*, tell what the child did wrong *and* write the right answer.



GOOD. No gaps and no overlaps.



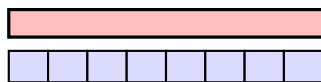
GAPS. Spaces are left, so too few cubes fit.



OVERLAPS. Cubes sit on top of each other, so too many fit.

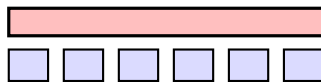
How cubes are laid down. No numbers are shown here — count the cubes in each problem's own picture.

1. Ivy laid her cubes under the red ribbon. There are **no gaps** and **no overlaps**. Count Ivy's cubes. How long is the ribbon?



Answer: _____ cubes

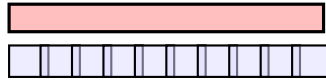
2. Max measured the **same** red ribbon. He left gaps, so he counted only 6 cubes. Ivy measured it carefully and got 8 cubes.



Write $<$ or $>$ in the box to tell the truth about Max's number:

Max's count real length

3. Find the mistake. Rex measured the **same** red ribbon. He let his cubes sit on top of each other, counted 10, and said “the ribbon is 10 cubes long.”



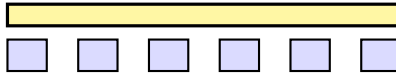
- (a) Tell Rex what he did wrong. (b) Write the ribbon's real length.

Answer: _____ cubes

4. Rex counted 10 cubes. Ivy counted 8 cubes. They measured the same ribbon. How many **more** cubes did Rex count than Ivy?

Answer: _____ cubes

5. Find the mistake. Sam measured a yellow straw. He left gaps between his cubes and counted 6. A careful measure shows the straw is really 10 cubes long.



(a) Tell Sam what he did wrong. (b) How many cubes did Sam's gaps hide?

Answer: _____ cubes

6. Rex says: "I counted 10 cubes and Ivy only counted 8, so my measuring is better."
Is Rex right? Draw a picture of the ribbon and the cubes, and tell how you know.